

Lecture2: Data

- Objects and their attributes
- Type of data
- Binning (Bucketization)
- The data mining pipeline
- Sampling
- Dimensionality Reduction
- Data Cleaning
- Deal with missing values

Lecture3: Data Grammar

- Normalization of numeric data
- What kind of normalization to choose?
- Common Pandas functions
- Similarity and Distance
 - (Cosine Similarity/Jaccard Similarity)
 - (Hamming Distance/ Variational distance/ Information theoretic distances/ Ranking distances/ Euclidean distance)
- Which similarity/ distance metric to choose

Lecture4: Get the Data

- How does a web server work?
- Why scrapes the web?
- Find elements by html class name
- Extract information from tags
- What is API

Lecture5: Data Visualization

- Visualization motivation
- Principle of Visualization
- Types of Visualization (What would you like to show)

Lecture6: Infrastructure that supports Big Data processing

- What is cluster architecture
- What is MapReduce (Map/Reduce process)
- Problems suited for MapReduce
- Design MapReduce functions (Identify input/output)
- Advantages in Spark
- Counting words example